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Ground compaction sensing system and method for a work machine

Abstract

A ground compaction sensing system for a work machine includes a chassis, a ground-engaging mechanism coupled to the chassis, and an attachment movably coupled to the chassis. Multiple sensors are configured to generate a chassis angle signal, an attachment angle signal, an attachment spacing signal, and a location signal. The controller has a non-transitory computer readable medium with a program instruction to grade a surface. The program instructions when executed cause a processor of the controller to receive the aforementioned signals and determine an as-built grade of the surface based on an attachment reference point at a first location of the location signal; and a compaction value when a chassis reference point reaches the first location as the work machine traverses across the surface. The program instructions may then include modifying movement of the attachment based on the compaction value.

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Background/Summary

TECHNICAL FIELD

(1) The disclosure generally relates to a ground compaction sensing system and method for a work machine.

BACKGROUND

(2) Grading operations with work machines is a specialized phase of the construction process. Proper ground preparation ensures expected outcomes in architectural construction, control of water runoff, road construction, environmental impact and compliance with land grading standards. Current technology enables fine-tuned global positioning systems to accurately track the location of surface creation from a design file with minimal direction from the operator. However, the as-built surfaces derived from the design file assume a 100% compaction of the ground surface, when in reality, many surface cutting and spreading operations may yield some loosely compacted material. This may require additional passes depending on the degree of compaction to meet specification. Alternatively, over compaction may lead to wasted time and fuel. Current methods include spot checking compaction throughout a worksite using an arduous manual process (e.g. using a cone penetrometer). Therein lies an opportunity for improved grading operations by accounting for other influences, such as vehicular traffic, in real-time.

SUMMARY

(3) A ground compaction sensing system and method for a work machine is disclosed. The ground compaction sensing system comprises a chassis, a ground-engaging mechanism coupled to the chassis; an attachment coupled to the chassis, a first sensor, a second sensor, a third sensor, a fourth sensor, and a controller. The first sensor is coupled to the chassis and configured to generate a chassis angle signal indicative of a chassis angle relative to the direction of gravity. The second sensor is coupled to the attachment and configured to generate an attachment angle signal indicative of an attachment angle relative to the direction of gravity. The third sensor is configured to generate an attachment spacing signal wherein a distance between the attachment and the chassis may be derived. The fourth sensor is configured to generate a location signal indicative of a location of one of the chassis and the attachment. The controller has a non-transitory computer readable medium with a program instruction to grade a surface. The program instructions when executed cause a processor of the controller to perform the following steps. In a first step, a processor will receive the chassis angle signal from the first sensor; receive the attachment angle from the second sensor; receive the attachment spacing signal from the third sensor and receive the location signal from the fourth sensor. Next, the processor determines an as-built surface based on the chassis angle signal, the attachment angle signal, the attachment spacing signal when a chassis reference point reaches the first location as the work machine traverses across the surface. Then, the processor modifies movement of the attachment based on the compaction value.

(4) The chassis reference point may be located on one of an aft portion of the chassis and behind the ground-engaging mechanism.

(5) The compaction value may represent an elevational difference of the as-built surface compacted by the ground engaging-mechanism.

(6) The processor may further control a display device to display one of the compaction value and an elevational difference in real-time as a singular graphic with a compacted surface overlayed with the as-built surface.

(7) The processor may further communicate the compaction value to a follower work machine wherein the follower work machine modifies a ground-engaging attachment coupled to the follower work machine in response to receiving the compaction value.

(8) Modifying movement of the attachment may comprise of adjusting the pitch of the attachment to reflect a desired grade.

(9) The processor may further generate a ground compaction stress map indicative of the compaction values across a worksite, store the ground compaction stress map in a memory, and

identify a compaction impact on a subsequent of grading from one or more work machines.

(10) The processor may further access the ground compaction stress map, modify movement of one of the attachment and an alternative attachment, based on the compaction impact from the ground compaction stress map.

(11) The method of sensing ground compaction from a work machine comprises of the following. In a first step, the method includes generating a chassis angle signal from a first sensor coupled to the chassis of the work machine wherein the chassis angle signal is indicative of the chassis angle relative to a direction of gravity. The method also includes generating an attachment spacing signal from a third sensor coupled to the work machine wherein the distance between the attachment and the chassis may be derived. Then, the method includes generating a location signal from a fourth sensor coupled to the work machine wherein the location signal is indicative of either the chassis or the attachment.

(12) In a next step, the method includes receiving the chassis angle signal from the first sensor by a controller of the work machine, receiving the attachment angle signal from the second sensor by the controller, receiving the attachment spacing signal from the third sensor by the controller, and receiving the location signal from the fourth sensor by the controller. Subsequently, the method includes determining an as-built grade of a surface based on an attachment reference at a first location of the location signal, and then determining a compaction value of the as-built surface. The compaction value is based on the chassis angle signal, the attachment angle signal, the attachment spacing signal, and the location signal when a chassis reference point reaches the first location as the work machine traverses across the surface. Finally, the method includes modifying movement of the work machine based on the compaction value.

(13) The method may further comprise communicating the compaction value to a follower work machine wherein the follower work machine modifies a ground-engaging attachment coupled to the follower work machine in response to receiving the compaction value.

(14) The method may further comprise generating a ground compaction stress map indicative of the compaction value across a worksite, store the ground the compaction stress map in a memory, and identify a compaction impact on a subsequent pass of grading from one or more work machines.

(15) The method may further comprise accessing the ground compaction stress map, modifying movement of one of the attachment and an alternative attachment wherein the alternative attachment is coupled to an alternative work machine. The modification occurs based on the compaction impact of a location from the ground compaction map.

(16) The above features and advantages and other features and advantages of the present teachings are readily apparent from the following detailed description of the best modes for carrying out the teachings when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of one embodiment of a work machine, shown as a crawler dozer.

(2) FIG. 2 is a block diagram of one embodiment of the system architecture and the flow of the ground compaction sensing system.

(3) FIG. 3 is an illustration depicting a work surface during a grading operation with the work machine.

(4) FIG. 4 is an illustration depicting a work surface during a grading operation with a follower work machine.

(5) FIG. 5 is an illustration of a worksite using the ground compaction sensing system.

(6) FIG. 6a is an exemplary embodiment of a display device showing a compaction value in real-time.

(7) FIG. 6*b* is an exemplary embodiment of a display device showing a ground compaction stress map.

(8) FIG. 7 is a method of sensing ground compaction with a work machine using grade control according to a first embodiment.

(9) FIG. 8 is a logic flow diagram of the ground compaction sensing program instructions.

DETAILED DESCRIPTION

(10) Those having ordinary skill in the art will recognize that terms such as “above,” “below,” “upward,” “downward,” “top,” “bottom,” etc., are used descriptively for the figures, and do not represent limitations on the scope of the disclosure, as defined by the appended claims.

Furthermore, the teachings may be described herein in terms of functional and/or logical block components and/or various processing steps. It should be realized that such block components may be comprised of any number of hardware, software, and/or firmware components configured to perform the specified functions.

(11) Terms of degree, such as “generally”, “substantially” or “approximately” are understood by those of ordinary skill to refer to reasonable ranges outside of a given value or orientation, for example, general tolerances or positional relationships associated with manufacturing, assembly, and use of the described embodiments.

(12) In addition, as used herein, the term “or” is an inclusive “or” operator and is equivalent to the term “and/or,” unless the context clearly dictates otherwise. The term “based on” is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise.

(13) As used herein, unless otherwise limited or modified, lists with elements that are separated by conjunctive terms (e.g., “and”) and that are also preceded by the phrase “one or more of” or “at least one of” indicate configurations or arrangements that potentially include individual elements of the list, or any combination thereof. For example, “at least one of A, B, and C” or “one or more of A, B, and C” indicates the possibilities of only A, only B, only C, or any combination of two or more of A, B, and C (e.g., A and B; B and C; A and C; or A, B, and C).

(14) As used herein, “controller” **10** is intended to be used consistent with how the term is used by a person of skill in the art, and refers to a computing component with processing, memory **20**, and communication capabilities, which is utilized to execute instructions (i.e., stored on the memory **20** or received via the communication capabilities) to control or communicate with one or more other components. In certain embodiments, the controller **10** may be configured to receive input signals in various formats (e.g., hydraulic signals, voltage signals, current signals, CAN messages, optical signals, radio signals), and to output command or communication signals in various formats (e.g., hydraulic signals, voltage signals, current signals, CAN messages, optical signals, radio signals).

(15) The controller **10** may be in communication with other components on the work machine **100**, such as hydraulic components, electrical components, and operator inputs within an operator station of an associated work machine. The controller **10** may be electrically connected to these other components by a wiring harness such that messages, commands, and electrical power may be transmitted between the controller **10** and the other components. Although the controller **10** is referenced in the singular, in alternative embodiments the configuration and functionality described herein can be split across multiple devices using techniques known to a person of ordinary skill in the art. The controller **10** includes the tangible, non-transitory memory **20** on which are recorded computer-executable instructions, including a ground compaction sensing program instructions **40**. The processor **30** of the controller **10** is configured for executing the ground compaction sensing program instructions **40**.

(16) The controller **10** may be embodied as one or multiple digital computers or host machines each having one or more processors, read only memory (ROM), random access memory (RAM), electrically-programmable read only memory (EPROM), optical drives, magnetic drives, etc., a high-speed clock, analog-to-digital (A/D) circuitry, digital-to-analog (D/A) circuitry, and any

required input/output (I/O) circuitry, I/O devices, and communication interfaces, as well as signal conditioning and buffer electronics.

(17) The computer-readable memory may include any non-transitory/tangible medium which participates in providing data or computer-readable instructions. The memory **20** may be non-volatile or volatile. Non-volatile media may include, for example, optical or magnetic disks and other persistent memory. Example volatile media may include dynamic random-access memory (DRAM), which may constitute a main memory. Other examples of embodiments for memory **20** include a floppy, flexible disk, or hard disk, magnetic tape or other magnetic medium, a CD-ROM, DVD, and/or any other optical medium, as well as other possible memory devices such as flash memory.

(18) As such, a method **800** may be embodied as a program or algorithm operable on a controller **10**. It should be appreciated that the controller **10** may include any device capable of analyzing data from various sensors, comparing data, making decisions, and executing the required tasks.

(19) FIG. **1** is a perspective view of work vehicle **100**. Work machine **100** is illustrated as a crawler dozer, which may also be referred to as a crawler, but may be any work machine with a ground-engaging blade or work implement such as a compact track loader, motor grader, scraper, skid steer, and tractor, to name a few examples. Work machine **100** may be operated to engage the ground and cut and move material to achieve simple or complex features on the ground. As used herein, directions with regard to work machine **100** may be referred to from the perspective of an operator seated within operator station **136**: the left of work machine **100** is to the left of such an operator, the right of work machine **100** is to the right of such an operator, the front or fore of work machine **100** is the direction such an operator faces, the rear or aft of work machine **100** is behind such an operator, the top of work machine **100** is above such an operator, and the bottom of work machine **100** is below such an operator. While operating, work machine **100** may experience movement in three directions and rotation in three directions. Direction for work machine **100** may also be referred to with regard to longitude **102** or the longitudinal direction, latitude **106** or the lateral direction, and vertical **110** or the vertical direction. Rotation for work machine **100** may be referred to as roll **104** or the roll direction, pitch **108** or the pitch direction, and yaw **112** or the yaw direction or heading.

(20) Work machine **100** is supported on the ground by chassis **114**. Chassis **114** includes left track **116** and right track **118**, which engage the ground and provide tractive force for work machine **100**. Left track **116** and right track **118** may be comprised of shoes with grousers that sink into the ground to increase traction, and interconnecting components that allow the tracks to rotate about front idlers **120**, track rollers **122**, rear sprockets **124** and top idlers **126**. Such interconnecting components may include links, pins, bushings, and guides, to name a few components. Front idlers **120**, track rollers **122**, and rear sprockets **124**, on both the left and right sides of work machine **100**, provide support for work machine **100** on the ground. Front idlers **120**, track rollers **122**, rear sprockets **124**, and top idlers **126** are all pivotally connected to the remainder of work machine **100** and rotationally coupled to their respective tracks so as to rotate with those tracks. Track portion **128** of the chassis **114** provides structural support or strength to these components and the remainder of chassis **114**.

(21) Front idlers **120** are positioned at the longitudinal front of left track **116** and right track **118** and provide a rotating surface for the tracks to rotate about and a support point to transfer force between work machine **100** and the ground. Left track **116** and right track **118** rotate about front idlers **120** as they transition between their vertically lower and vertically upper portions parallel to the ground, so approximately half of the outer diameter of each of front idlers **120** is engaged with left track **116** or right track **118**. This engagement may be through a sprocket and pin arrangement, where pins included in left track **116** and right track **118** are engaged by recesses in front idler **120** so as to transfer force. This engagement also results in the vertical height of left track **116** and right track **118** being only slightly larger than the outer diameter of each of front idlers **120** at the

longitudinal front of left track **116** and right track **118**. Frontmost engaging point **130** of left track **116** and right track **118** can be approximated as the point on each track vertically below the center of front idlers **120**, which is the frontmost point of left track **116** and right track **118** which engages the ground. When work machine **100** encounters a ground feature when traveling in a forward direction, left track **116** and right track **118** may first encounter it at frontmost engaging point **130**. If the ground feature is at a higher elevation than the surrounding ground surface (i.e., an upward ground feature), work machine **100** may begin pitching backward (which may also be referred to as pitching upward) when frontmost engaging point **130** reaches the ground feature. If the ground feature is at a lower elevation than the surrounding ground surface (i.e., a downward ground feature), work machine **100** may continue forward without pitching until the center of gravity of work machine **100** is vertically above the edge of the downward ground feature. At that point, work machine **100** may pitch forward (which may also be referred to as pitching downward) until frontmost engaging point **130** contacts the ground. In this embodiment, front idlers **120** are not powered and thus are freely driven by left track **116** and right track **118**. In alternative embodiments, front idlers **120** may be powered, such as by an electric or hydraulic motor, or may have an included braking mechanism configured to resist rotation and thereby slow the left track **116** and right track **118**.

(22) Track rollers **122** are longitudinally positioned between front idler **120** and rear sprocket **124** along the bottom left and bottom right sides of work machine **100**. Each of track rollers **122** may be rotationally coupled to left track **116** or right track **118** through engagement between an upper surface of the tracks and a lower surface of track rollers **122**. This configuration may allow track rollers **122** to provide support to work machine **100**, and in particular may allow for the transfer of forces in the vertical direction between work machine **100** and the ground. This configuration also resists the upward deflection of left track **116** and right track **118** as they traverse an upward ground feature whose longitudinal length is less than the distance between front idler **120** and rear sprocket **124**.

(23) Rear sprockets **124** may be positioned at the longitudinal rear of left track **116** and right track **118** and, similar to front idlers **120**, provide a rotating surface for the tracks to rotate about and a support point to transfer force between work machine **100** and the ground. Left track **116** and right track **118** rotate about rear sprockets **124** as they transition between their vertically lower and vertically upper portions parallel to the ground, so approximately half of the outer diameter of each of rear sprockets **124** is engaged with left track **116** or right track **118**. This engagement may be through a sprocket and pin arrangement, where pins included in left track **116** and right track **118** are engaged by recesses in rear sprockets **124** so as to transfer force. This engagement also results in the vertical height of left track **116** and right track **118** being only slightly larger than the outer diameter of each of rear sprockets **124** at the longitudinal back or rear of left track **116** and right track **118**. Rearmost engaging point **132** of left track **116** and right track **118** can be approximated as the point on each track vertically below the center of rear sprockets **124**, which is the rearmost point of left track **116** and right track **118** which engages the ground.

(24) In this embodiment, each of rear sprockets **124** may be powered by a rotationally coupled hydraulic motor so as to drive the left track **116** and right track **118** and thereby control propulsion and traction for work machine **100**. Each of the left and right hydraulic motors may receive pressurized hydraulic fluid from a hydrostatic pump **138** whose direction of flow and displacement controls the direction of rotation and speed of rotation for the left and right hydraulic motors. Each hydrostatic pump **138** may be driven by engine **134** of work machine **100** and may be controlled by an operator in operator station **136** issuing commands which may be received by controller **10** and communicated to the left and right hydrostatic pumps **138** by controller **10**. In alternative embodiments, each of rear sprockets **124** may be driven by a rotationally coupled electric motor or a mechanical system transmitting power from engine **134**.

(25) Top idlers **126** are longitudinally positioned between front idlers **120** and rear sprockets **124**

along the left and right sides of work vehicle **100** above track rollers **122**. Similar to track rollers **122**, each of top idlers **126** may be rotationally coupled to left track **116** or right track **118** through engagement between a lower surface of the tracks and an upper surface of top idlers **126**. This configuration may allow top idlers **126** to support left track **116** and right track **118** for the longitudinal span between front idler **120** and rear sprocket **124** and prevent downward deflection of the upper portion of left track **116** and right track **118** parallel to the ground between front idler **120** and rear sprocket **124**.

(26) A first sensor **144** is affixed to chassis **140** of work vehicle **100** and configured to provide a chassis angle signal **202** indicative of the movement and orientation of chassis **140**. In alternative embodiments, first sensor **144** may not be affixed directly to chassis **140** but may instead be connected to chassis **140** through intermediate components or structures, such as rubberized mounts. Connecting the first sensor **144** to chassis **140** in a fixed relative position through the use of mounts or brackets may allow first sensor **144** to experience and measure the motion of chassis **140**, enabling measurements by first sensor **144** to be indicative of the similar measurements taken from the first sensor directly affixed to chassis **140**.

(27) The first sensor **144** is a component configured to provide a signal indicative of the angle of chassis **140** in the direction of roll **104** and the angular velocity of chassis **140** in the direction of roll **104**. For example, these signals may be referred to as a chassis tilt signal and a chassis roll signal, respectively. First sensor **144** may also be configured to provide a signal or signals indicative of other positions or velocities of chassis **140**, including its inclination (i.e., an angle of chassis **140** relative to the direction of gravity) in a direction such as the direction of roll **104**, pitch **108**, and yaw **112**, its angular velocity or angular acceleration in a direction such as the direction of roll **104**, pitch **108**, yaw **112**, or its linear velocity or linear acceleration in a direction such as the direction of longitude **102**, latitude **106**, and vertical **110**. These may collectively be referred to chassis angle signal **202**. First sensor **144** may be configured to directly measure angular acceleration, angular velocity, or angular position, or measure one of these and derive or integrate the measurements to arrive at another of these (e.g., integrate angular velocity to arrive at angular position).

(28) Attachment **142** may be a blade or other type of tool which may engage the ground or material to move or shape it. Attachment **142** may be used to move material from one location to another to create features on the ground, including flat areas, grades, hills, roads, or more complexly shaped features. In this embodiment, attachment **142** of work vehicle **100** may hereinafter be referred to as a blade, a six-way blade, six-way adjustable blade, or power-angle-tilt (PAT) blade. Attachment **142** may be hydraulically actuated to move vertically up or vertically down (which may also be referred to as lift, or raise and lower), roll left or roll right (which may be referred to as tilt, or tilt left and tilt right), and yaw left or yaw right (which may be referred to as blade angle, or angle left and angle right). Alternative embodiments may utilize an attachment with fewer hydraulically controlled degrees of freedom, such as a 4-way attachment that may not be angled or actuated in the direction of yaw **112**.

(29) Attachment **142** is movably coupled to the chassis **140** of work machine **100** through linkage **146**, which supports and actuates attachment **142** and is configured to allow attachment **142** to be tilted relative to chassis **140** (i.e., moved in the direction of roll **104**). Linkage **146** may include multiple structural members to carry forces between blade **142** and the remainder of work vehicle **100** and may provide attachment points for hydraulic cylinders which may actuate the attachment in the lift, tilt, and angle directions.

(30) In the present embodiment, linkage **146** includes a c-chassis **148**, a structural member with a C-shape positioned rearward of blade **142**, with the C-shape open toward the rear of work vehicle **100**. Each rearward end of c-chassis **148** is pivotally connected to chassis **140** of work vehicle **100**, such as through a pin-bushing joint, allowing the front of c-chassis **148** to be raised or lowered relative to work vehicle **100** about the pivotal connections at the rear of c-chassis **148**. The front

portion of c-chassis **148**, which is approximately positioned at the lateral center of work vehicle **100**, connects to blade **142** through a ball-socket joint. This allows blade **142** three degrees of freedom in its orientation relative to c-chassis **148** (lift-tilt-angle) while still transferring rearward forces on blade **142** to the remainder of work vehicle **100**.

(31) Now referring to FIGS. 2-4, a ground compaction sensing system **200** for a work machine **100** assists in grading operations using an attachment **142**. The ground compaction sensing system **200** may comprise of the chassis **140**, the ground-engaging mechanism **150** coupled to the chassis **140**, and a multitude of sensors coupled to the controller **10**. The first sensor **144** is coupled to the chassis **140** and configured to generate a chassis angle signal **202** indicative of a chassis angle **204** relative to a direction of gravity.

(32) A second sensor **244**, functionally similar to the first sensor **144**, is coupled to the attachment **142** wherein the second sensor **244** is configured to generate an attachment angle signal **212** indicative of an attachment angle **214** relative to the direction of gravity. The second sensor **244** is configured to directly measure angular acceleration, angular velocity, or angular position, or measure one of these and derive or integrate the measurements to arrive at another of these (e.g., integrate angular velocity to arrive at angular position). The second sensor **244** may be affixed to blade **142** above the ball-socket joint connecting blade **142** to c-chassis **148**. The second sensor **244**, like the first sensor **144**, may be configured to measure orientation, angular velocity, or acceleration. Second sensor **244** may be connected to blade **142** through an intermediate component, such as a bracket, mount, or portion of linkage **146**, at a fixed relative position to blade **142** so that may experience and measure the motion of blade **142**, enabling measurements by second sensor **244** to be indicative of similar measurements taken from a sensor directly affixed to blade **142**. Second sensor **244** may include one more gyroscopes which it may use to sense angular velocities and one or more accelerometers which it may use to measure linear acceleration. Second sensor **244** may sense the tilt angle of blade **142** by measuring linear acceleration in three substantially perpendicular axes, and using those measurements to determine the direction of gravity and thereby determine the tilt angle of blade **142**.

(33) Blade **142** may be raised or lowered relative to work vehicle **100** by the actuation of lift cylinders, which may raise and lower c-chassis **148** and thus raise and lower blade **142**, which may also be referred to as blade lift. For each of lift cylinders, the rod end is pivotally connected to an upward projecting clevis of c-chassis **148** and the head end is pivotally connected to the remainder of work vehicle **100** just below and forward of operator station **136**. The configuration of linkage **146** and the positioning of the pivotal connections for the head end and rod end of lift cylinders results in the extension of lift cylinders lowering blade **142** and the retraction of lift cylinders raising blade **142**. In alternative embodiments, blade **142** may be raised or lowered by a different mechanism, or lift cylinders may be configured differently, such as a configuration in which the extension of lift cylinders raises blade **142** and the retraction of lift cylinders lowers blade **142**.

(34) Blade **142** may be tilted relative to work vehicle **100** by the actuation of tilt cylinder, which may also be referred to as moving blade **142** in the direction of roll **104**. For tilt cylinder, the rod end is pivotally connected to a clevis positioned on the back and left sides of blade **142** above the ball-socket joint between blade **142** and c-chassis **148** and the head end is pivotally connected to an upward projecting portion of linkage **146**. The positioning of the pivotal connections for the head end and the rod end of tilt cylinder result in the extension of tilt cylinder tilting blade **142** to the left or counterclockwise when viewed from operator station **136** and the retraction of tilt cylinder tilting blade **142** to the right or clockwise when viewed from operator station **136**. In alternative embodiments, blade **142** may be tilted by a different mechanism (e.g., an electrical or hydraulic motor) or tilt cylinder may be configured differently, such as a configuration in which it is mounted vertically and positioned on the left or right side of blade **142**, or a configuration with two tilt cylinders.

(35) Blade **142** may be angled relative to work vehicle **100** by the actuation of angle cylinders **154**,

which may also be referred to as moving blade **142** in the direction of yaw **112**. For each of angle cylinders **154**, the rod end is pivotally connected to a clevis of blade **142** while the head end is pivotally connected to a clevis of c-chassis **148**. One of the angle cylinders **154** is positioned on the left side of work vehicle **100**, left of the ball-socket joint between blade **142** and c-chassis **148**, and the other of the angle cylinders **154** is positioned on the right side of work vehicle **100**, right of the ball-socket joint between blade **142** and c-chassis **148**. This positioning results in the extension of the left of angle cylinders **154** and the retraction of the right of angle cylinders **154** angling blade **142** rightward, or yawing blade **142** clockwise when viewed from above, and the retraction of left of angle cylinder **154** and the extension of the right of angle cylinders **154** angling blade **142** leftward, or yawing blade **142** counterclockwise when viewed from above. In alternative embodiments, blade **142** may be angled by a different mechanism or angle cylinders **154** may be configured differently.

(36) A third sensor **254** is configured to generate an attachment spacing signal **256** wherein a distance between the attachment **142** and the chassis **140** may be derived. More specifically, the third sensor **254** senses, in the fore-aft direction, one of the spacing between a first point on the chassis **140** relative to a second point on the attachment and a change in spacing therebetween, as the attachment is moved. The third sensor **254** is configured to directly measure the spacing, or can be defined as a derived spacing or change in spacing through pre-existing sensors such as hydraulic sensors related to linkage kinematics.

(37) A fourth sensor **264** is configured to generate a location signal **266** indicative of a location of one of the chassis **140** and the attachment **142**. In one embodiment, the fourth sensor **264** derive a location of the work machine from one or more of a base station and a global navigation satellite system (GNSS) receiver. The GNSS receiver can receive and compute its position from the signals provided by the global navigation satellites. The work machine **100**, in addition to knowing its own position (as computed from the detected satellite signals received), may also be provided with a location signal **266** from a base station at a known and fixed position. The controller **10** has a non-transitory computer readable medium with a program instruction **40** to grade the surface **50**. The program instructions **40** when executed cause a processor **30** of the controller **10** to perform the following. The processor **30** receives the chassis angle signal **202** from the first sensor **144**; the attachment angle signal **212** from the second sensor **244**; the attachment spacing signal **256** from the third sensor **254**; and a location signal **265** from the fourth sensor **264**.

(38) A chassis reference point **306** is located on an aft portion **308** of the chassis **140** and proximal to the ground-engaging mechanism **150**. In the embodiment shown in FIG. 3, the attachment reference point **302** may be located on or around the ball joint (for this particular embodiment) behind the attachment **142**. In other embodiments, the attachment reference point **302** may be on or proximal to the attachment **142** such as an attachment coupler, or a linkage.

(39) The program instructions **40** for the processor **30** will determine an as-built surface **152** (may also be referred to as an as-built grade) of the virgin surface **151** based on the attachment reference point **302** at a first location **304** of the location signal **266**. As used herein, “based on” means “based at least in part on” and does not mean “based solely on,” such that it neither excludes nor requires additional factors. As shown in FIG. 4, the work machine **100** is moving in the direction of the arrow. A compaction value **270** represents an elevational difference **310** of the as-built surface **152** and the as-built surface compacted **153** by the ground-engaging mechanism **150**. The schematic of the work machine in dotted lines is the work machine at t.sub.1. The schematic of the work machine in solid lines is the work machine at t.sub.2. FIG. 5 identifies a first as-built compacted **153a** by loader further compacted by a follower work machine **410** with a roller attachment **415** to create a second as-built compacted surface **153b**.

(40) The compaction value **270** of the as-built surface **152** is based on or derived from the chassis angle signal **202**, the attachment angle signal **212**, and the attachment spacing signal **256** when the chassis reference point **306** at t.sub.1 reaches the first location **304** (i.e. where the attachment

reference point **302** was at t.sub.1) at a later time t.sub.2 as the work machine **100** traverses across the surface **50**. The compaction value **270** is tracked in real-time as it can be calculated in seconds. In response, the program instructions **40** may cause a processor **30** on the controller **10** to modify movement of the attachment **142** based upon the compaction value **270**. Note the attachment reference point **302** and the chassis reference point **306** need not be in the same plane to calculate the compaction value **270**.

(41) Now turning to FIG. **6A**, the program instructions **40** when executed further causes the processor **30** of the controller **10** to control a display device **272** to display a real-time elevational difference **310** as a singular graphic **405** with a compacted surface **153** (shown in a bold line) overlayed with the as-built surface **152**. The grade profile of the as built surface **152** attachment **142** (e.g. cross slope and mainfall), in one embodiment, may be derived from receiving localized attachment corner positions wherein the attachment **142** is a blade. The program instructions **40** when executed further causes the processor **30** of the controller **10** to control a display device **272** to display the compaction value **270**. The compaction value **270** may be represented as a nominal value, a percentage of the desired grade, an in/out of tolerance indicator, or a number of passes remaining based on the true value **271** (i.e. the surface grade post compaction).

(42) The program instructions **40** when executed may further cause the processor **30** of the controller **10** to communicate the compaction value **270** to a follower work machine **410** (hereinafter also referred to as “alternative work machine”) wherein the follower work machine **410** adjusts the ground-engaging attachment **142** coupled to the follower work machine **410** in response to receiving the compaction value **270**. In industrial applications, compaction of large deposits of soil is required in road building wherein some areas may require artificial fills and others may require embankments. Compaction can be especially important when forming the foundation under bridge piers, buildings, roads, dams, levees, airports, and parks, e.g. Furthermore, the process can require multiple layers of materials such as the sub-grade, a drainable aggregate rock base, and then an asphalt base. An initial pass may be performed by a first work machine **100** with a blade type attachment, and a follower work machine **410** may follow with a drum type attachment for further compaction. The program instructions **40** for acquiring the true value **271** (i.e. the grade post compaction) or compaction value **270** may iteratively repeat with each pass, or with each follower work machine **410**, and thereby advantageously avoiding a potential cumulative error in grading operations, improve accuracy, and create a more uniform compaction of the worksite. Referring to FIG. **6B**, a well-understood map of compaction over an entire worksite (i.e. ground compaction stress map **275**) may advantageously enable restructuring of the time allotted from several difference work machines to specific areas. Use of compaction knowledge from the work machines can improve material handling practices by proactively leaving more material in low compaction areas during subsequent passes. For example, in road building, a worksite with fewer low compaction areas can reduce the aggregate required for filing in the next layer, and thereby aggregate material costs.

(43) Additionally, the program instructions **40** when executed may cause the processor **30** of the controller **10** to generate a ground compaction stress map **275** (shown in FIG. **6B**) indicative of the compaction values **270** across a worksite **280**; store the ground compaction stress map **275** in a memory **20**; and identify a compaction impact **420** on a subsequent pass of grading from one or more work machines. The ground compaction stress map **275** can indicate the degree of elevational differences **310** within a worksite and displayed as color-coded overlay of a worksite map. These values may be initially established during an initial pass of a given work area. In one embodiment, the ground compaction stress map **275** can be calibrated with a few manual soil-compaction tests at specific locations and identify the relationship between machine weight and soil compaction, as soil compaction may be machine specific. Calibrating may also automatically compensate for moisture and soil type, and thereby further improving the accuracy. The ground compaction sensing system integrated with known grade control systems provides a more accurate assessment, and

further provides this information in real-time.

(44) The program instructions **40** when executed may further cause the processor **30** of the controller **10** to access the ground compaction stress map **275**; modifying movement of one of the attachment **142** in a subsequent pass, or an alternative attachment **410** from one or more alternative work machines **410**. Potential improvements to grade control functions can be used the in creates understanding of compaction. For example, an accurate elevation difference between in real-time of the as-built surface **152** and the compacted surface **153** can inform the operator to adjusts the amount of material flowing under the blade attachment **142** by altering the attachment position and angle. The improved accuracy can further enable the work machine to adjust the attachment to perform a compaction and finishing elevation simultaneously.

(45) FIG. 7 is one embodiment of a logic flow diagram **700** of the program instructions **40** for the ground compaction sensing system **200**. The system **200** includes a series of processing instructions or steps from the controller **10** that are depicted in flow diagram form. The process begins with the controller **10** receiving the work machine position in step **705**. This includes receiving both the chassis angle signal **202** and the location signal. In step **710**, the controller **10** receives the attachment angle signal through identifying the localized blade corner positions when the attachment is blade. Positioning of the attachment relative to the chassis **140** may also be derived from the attachment angle signal. In step **715**, the controller **10** engages a compaction map. Subsequently in step **720**, the controller **10** compares the work machine position to a previously recorded blade path. The controller **10** then determines if the work machine is over an as-built surface in step **725**. If yes, then the controller **10** will subtract the compacted surface value with the as-built surface value to acquire the elevational difference in step **730**. If not, the controller **10** will continue checking its location with the previously recorded blade path. In step **735**, the controller **10** will log the elevation difference with a location and time stamp. In step **740**, the controller **10** will apply the compaction values **270** in relation to a map of the worksite wherein “red” may indicate greater compaction, and green may indicate “less” compaction. Based on the location compaction value measurements in step **745**, the controller **10** can alter the map to display whether the true grade is within or out of tolerance in step **750a**. If localized compaction values **270** are not acquired, the controller **10** may prompt the user **750b** to activate applying the relative compaction values to the map.

(46) FIG. 8 discloses a method **800** of sensing ground compaction with a work machine **100** using grade control according to one embodiment. The method comprises, in step **805a**, generating a chassis angle signal from a first sensor coupled to the chassis **140** of the work machine **100**. The chassis angle signal is indicative of a chassis angle relative to the direction of gravity. In step **805b**, the method includes generating an attachment angle signal from a second sensor coupled to an attachment of the work machine **100**. The attachment angle signal is indicative of the attachment angle relative to the direction of gravity. In step **805c**, the method includes generating an attachment spacing signal from a third sensor coupled to the work machine **100**, wherein a distance between the attachment and the chassis **140** may be derived. In step **805d**, the method includes generating a location signal **266** from a fourth sensor coupled to the work machine **100** wherein the location signal **266** is indicative of one of the chassis **140** and the attachment. In step **810**, the method includes receiving the chassis angle signal, the attachment angle signal, the attachment spacing signal, and the location signal **266**. Step **815** includes determining an as-built grade of the surface. Step **820** includes determining a compaction value of the as-built surface. Step **825** includes modifying movement of the work machine **100** based on compaction value **270**. In step **830a**, the method includes communicating the location specific compaction values **270** to one or more follower work machines **410** and modifying movement of the follower work machine **410** based on the localized compaction values **270** in step **835**. Additionally, the method in step **830b** can include controlling a display device to display one of the compaction value **270** and a real-time elevational difference as a singular graphic with a compacted surface **153** overlaid with the as-

built surface. Alternatively, in step 830c, the method includes generating a ground compaction stress map indicative of the compaction values 270 across a worksite, storing the ground compaction stress map in memory 20, and identifying the compaction impact on a subsequent pass of grading from one or more work machines (100, 410).

(47) The ground compaction sensing system and method for a work machine with grade control disclosed herein has certain advantages. Notably, the system can maintain accuracy and continuity of the grading operation, and thereby eliminating inefficiencies in the process. Furthermore, the system enables a work machine 100 to run automated by self-correcting in equipment function, without necessarily requiring an operator to be present in the work machine 100.

(48) As used herein, “e.g.” is utilized to non-exhaustively list examples, and carries the same meaning as alternative illustrative phrases such as “including,” “including, but not limited to,” and “including without limitation.” As used herein, unless otherwise limited or modified, lists with elements that are separated by conjunctive terms (e.g., “and”) and that are also preceded by the phrase “one or more of,” “at least one of,” “at least,” or a like phrase, indicate configurations or arrangements that potentially include individual elements of the list, or any combination thereof. For example, “at least one of A, B, and C” and “one or more of A, B, and C” each indicate the possibility of only A, only B, only C, or any combination of two or more of A, B, and C (A and B; A and C; B and C; or A, B, and C). As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Further, “comprises,” “includes,” and like phrases are intended to specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

Claims

1. A ground compaction sensing system for a work machine, the work machine extending in a fore-aft direction, the ground compaction sensing system comprising: a chassis; a ground-engaging mechanism coupled to the chassis; a grading attachment movably coupled to the chassis; a first sensor coupled to the chassis, the first sensor configured to generate a chassis angle signal indicative of a chassis angle relative to a direction of gravity; a second sensor coupled to the grading attachment, the second sensor configured to generate a grading attachment angle signal indicative of an attachment angle relative to the direction of gravity; a third sensor configured to generate an attachment spacing signal wherein a distance between the attachment and the chassis is derived; a fourth sensor configured to generate a location signal indicative of a location of one of the chassis and the grading attachment; and a controller having a non-transitory computer readable medium with a program instruction to grade a surface, the program instructions when executed causing a processor of the controller to: receive the chassis angle signal from the first sensor; receive the attachment angle signal from the second sensor; receive the attachment spacing signal from the third sensor; receive the location signal from the fourth sensor; determining an as-built grade of the surface based on an attachment reference point at a first location of the location signal; determining a compaction value of the as-built surface based on the chassis angle signal, the attachment angle signal, and the attachment spacing signal when a chassis reference point reaches the first location as the work machine traverses across the surface, wherein the compaction value is an elevational difference of the as-built surface and the as-built surface compacted by the ground-engaging mechanism; and modifying movement of the grading attachment based on the compaction value.

2. The ground compaction sensing system of claim 1 wherein the chassis reference point is located on one of an aft portion of the chassis and behind the ground-engaging mechanism.

3. The ground compaction sensing system of claim 1 wherein the program instructions when executed further causes the processor of the controller to control a display device to display one of

the compaction value and an elevational difference in real-time as a singular graphic with a compacted surface overlayed with the as-built surface.

4. The ground compaction sensing system of claim 1 wherein the program instructions when executed further causes the processor of the controller to communicate the compaction value to a follower work machine wherein the follower work machine modifies a ground-engaging attachment coupled to the follower work machine in response to receiving the compaction value.

5. The ground compaction sensing system of claim 1 wherein the modifying movement of the grading attachment comprises adjusting a pitch of the grading attachment to reflect a desired grade.

6. The ground compaction sensing system of claim 1 wherein the program instructions when executed further causes the processor of the controller to generate a ground compaction stress map indicative of the compaction values across a worksite; store the ground compaction stress map in a memory; and identify a compaction impact on a subsequent pass of grading from one or more work machines.

7. The ground compaction sensing system of claim 6 wherein the program instructions when executed further causes the processor of the controller to access the ground compaction stress map; modifying movement of one of the grading attachment and an alternative attachment, the alternative attachment coupled to an alternative work machine, based on the compaction impact from the ground compaction stress map.

8. A method of sensing ground compaction from a work machine, the work machine extending in a fore-aft direction, the method comprising: generating a chassis angle signal from a first sensor coupled to a chassis of the work machine, the chassis angle signal indicative of a chassis angle relative to a direction of gravity; generating an attachment angle signal from a second sensor coupled to an attachment of the work machine, the attachment angle signal indicative of an attachment angle relative to the direction of gravity; generating an attachment spacing signal from a third sensor coupled to the work machine wherein a distance between a grading attachment and the chassis is derived; generating a location signal from a fourth sensor coupled to the work machine, the location signal indicative of one of the chassis and the attachment; receiving the chassis angle signal from the first sensor by a controller of the work machine; receiving the attachment angle signal from the second sensor by the controller; receiving the attachment spacing signal from the third sensor by the controller; receiving the location signal from the fourth sensor by the controller; determining an as-built grade of a surface based on an attachment reference at a first location of the location signal; determining a compaction value of the as-built surface based in the chassis angle signal, the attachment angle signal, the attachment spacing signal, and the location signal when a chassis reference point reaches the first location as the work machine traverses across the surface wherein the compaction value is an elevational difference of the as-built surface and the as-built surface compacted by the ground-engaging mechanism; and modifying movement of the work machine based on the compaction value.

9. The method of sensing ground compaction of claim 8 wherein the chassis reference point is located on one of an aft portion of the chassis and behind the ground-engaging mechanism.

10. The method of sensing ground compaction of claim 8 further comprising controlling a display device to display one of the compaction value and the elevational difference in real-time as a singular graphic with a compacted surface overlayed with the as-built surface.

11. The method of sensing ground compaction of claim 8 further comprising communicating the compaction value to a follower work machine wherein the follower work machine modifies a ground-engaging attachment coupled to the follower work machine in response to receiving the compaction value.

12. The method of sensing ground compaction of claim 8 wherein the modifying movement of the grading attachment comprises adjusting a pitch of the grading attachment to reflect a desired grade.

13. The method of sensing ground compaction of claim 8 further comprising generating a ground compaction stress map indicative of the compaction values across a worksite; store the ground

compaction stress map in a memory; and identify a compaction impact on a subsequent pass of grading from one or more work machines.

14. The method of sensing grade of claim 13 further comprising accessing the ground compaction stress map; modifying movement of one of the grading attachment and an alternative attachment, the alternative attachment coupled to an alternative work machine, based on the compaction impact of a location from the ground compaction stress map.
